

(Single Inheritance)

Write a program contains a struct Course that has data members:

C_name (string), Degree (float), Credit (int).

It contains a class Level that has:

- i. Data members: L[5](Course).**
- ii. A function read() to read data members.**
- iii. A function Print() to display data members in tabular form.**

It contains a class Student which is a subclass from Level that has:

- i. Data member: name (string), ID(int), total_d(float), total_c(int), GPA(char).**
- ii. A function read() to read data members, and set the value of total_d which is equal to the sum of all degrees for all courses, total_c which is the sum of Credit for all courses, and set the value of GPA according to the following table:**

GPA	Average of degree %
A	≥ 92
B+	84: <92
B	80: <84
C+	75: <80
C	65: <75
D	60: <65
F	<60

- iii. A function Print() to display data members in one line.**

In main function, define an array of Student class with length n, read and print the elements of this array in suitable form for each object.

(Multiple Inheritance)

Write a program contains a class Data1 that has:

- i. Data members: S1 (string), D1 [20] (double), a(int), n (number of elements).
- ii. A function to read data members.
- iii. A function Sum() to return the following sum:

$$\sum_{i=0}^n \begin{cases} D1_i^2 & \text{if } D1_i \text{ is divisible by } a \\ D1_i^3 & \text{otherwise} \end{cases}$$

- iv. A function display () to display data members.

It contains a class Data2 that has:

- i. Data members: S2 (string), D2[20](double), b(int), m (number of elements).
- ii. A function to read data members.
- iii. A function Sum() to return the following sum:

$$\sum_{i=0}^m \begin{cases} 2D2_i^2 & \text{if } D2_i > b \\ 3D2_i^3 & \text{otherwise} \end{cases}$$

- iv. A function display () to display data members.

It contains a class Data is a subclass of Data1 and Data2 that has:

- i. Data members: ND (string), Sum_T(double).
- ii. A function to read ND and set the value of Sum_T that is the sum of the elements of D1 and the elements of D2 (use Sum () function for each class)
- iii. Function to return the average of all data members for D1 and D2.
- iv. A function display () to display data members.

In main function, define objects of Data class and apply all functions on them.

Multilevel Inheritance

Write a program contains A class Rectangle that has:

- i. Data members: Dim[10][2] (float), n (number of rows).**
- ii. A function to read data members.**
- iii. A function to return the area of rectangle with dimensions Dim_{a, 0}, Dim_{a,1} for given index a.**
- iv. A function Print() to display data members in table form.**

It contains a class RArea which is a subclass of Rectangle that has:

- i. Data member: RA[10] (float).**
- ii. A function to set the elements of RA such that each RA_i is the area which is computed from ith row in Dim matrix (i=0,..,n-1).**
- iii. A function Print() to display data members in a list form.**

It contains a class P_volum which is a subclass from class RArea that has:

- i. Data members: H[10] (float), V[10] (float).**
- ii. A function to set data members such that each element V_i is equal to $H_i \times RA_i$ for i=0...n-1.**
- iii. A function Print() to display data members in table form.**

In main function, define objects of P_volum class and apply all functions on them.